"CHIP DESIGN IS ONE OF THE MOST COMPLEX ENGINEERING DOMAINS IMAGINABLE, AND THERE IS HUGE OPPORTUNITY FOR AI—PARTICULARLY AN AGENTIC APPROACH—TO TRANSFORM EVERYTHING FROM LOGIC DESIGN TO BOARD VALIDATION."

— GOKUL V. SUBRAMANIAM, INTEL INDIA PRESIDENT (IN A RECENT INTERVIEW WITH THE TIMES OF INDIA)

analyse, and optimise network performance in real time. Embedded within network devices, these agents can detect anomalies, predict potential failures, and implement corrective actions proactively without human intervention. This autonomy allows networks to adapt dynamically to changing conditions and user demands, leading to enhanced reliability and performance," says Pramod Gummaraj, founder and CEO of Aprecomm.

- ***Engineering:*** Agentic AI can play a significant role in many cutting-edge engineering fields, like chip design. Electronic design automation (EDA) vendors like Synopsis, as well as chipmakers like Intel, have started trialling agentic AI. "Chip design is one of the most complex engineering domains imaginable, and there is huge opportunity for AI—particularly an agentic approach—to transform everything from logic design to board validation," said Gokul Subramaniam, Intel India president and VP of the client computing group at Intel, in a recent interview with the *Times of India.* Engineering teams at Intel India are developing AI systems to automate chip design.

Subramaniam explained that each stage in semiconductor engineering, such as design, development, testing, validation and deployment, has its own workflow with numerous tools and people involved. "We believe an agentic approach—thinking of them as sophisticated assistants—can help with tasks like generating code, running simulations, or assisting with placement (of transistors), all while engineers retain full oversight," he said.

Intel is not building LLMs from the ground up, but using tools like LangChain or LangGraph, and layering their domain-specific knowledge on top. They are training the agents using historical data collected at Intel over 50+ years. Subramaniam highlighted that their engineers are encouraged to document their insights, not just for their benefit but also to feed the AI agents.

He suggested that if the fabless companies mushrooming in India can adopt AI-driven workflows, they could get to market faster and with fewer design spins, which could significantly lower barriers to entry.

The path ahead

With its abilities to autonomously plan and execute the steps needed to achieve a goal, work in a chain of command, seamlessly integrate with existing systems and human employees, learn from its experiences, and constantly improve, agentic AI has proven to be a worthy investment in many organisations—a calm yet high-performing employee, to say the least. According to a recent report by the Boston Consulting Group, people must get used to working closely with AI agents as teammates. "AI agents will be onboarded, just like human workers, to learn roles and responsibilities, access relevant company data and business context, integrate into workflows, and support the humans' responsibilities," it says.

In a recent media interview, Stuart Brown, head of digital business at Guidehouse, a PwC spin-off that consults for commercial businesses and governments, said, "Many people don't understand the impact… Some still think it's just another tool. But agentic AI will bring a fundamental change in how we operate. It will create new ways of working."

Indeed, many companies have already deployed AI agents for various tasks, and many are taking the leap. But, take it all with a pinch of salt. AI experts and business leaders warn that while agentic AI has great potential to transform businesses, it is not a silver bullet for all enterprise needs. Like all technologies that leapfrogged to the mainstream, AI has its share of problems, all of which trickle down to agentic AI.

We read about AI hallucinations in the February 2025 issue of *Electronics For You.* AI agents are known to have fabricated non-existent clients, bank transactions, legal precedents and

"AI IS DOING A GREAT JOB ELIMINATING THE ACCIDENTAL COMPLEXITY. HUMANS ARE STILL NEEDED TO DEAL WITH THE ESSENTIAL COMPLEXITY. AI CAN MAKE MINCEMEAT OF PATTERNS ALREADY DISCOVERED (BY HUMANS). CAN IT FIND TOTALLY NEW PATTERNS… I DON'T KNOW IF AI CAN DO THIS. I DON'T KNOW IF THAT CAN BE BRUTE-FORCED."

— SRIDHAR VEMBU, FOUNDER & FORMER CEO, ZOHO CORPORATION (IN A POST ON X)